Seminar on Positivity in Algebraic Geometry

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1 The Classical Theory

Definition 1.1 (Ample and very ample line bundles and divisors on a complete scheme)**.** Let X be a complete scheme, and L a line bundle on X.

(i). L is **very ample** if there exists a closed embedding $X \subset \mathbb{P}$ of X into some projective space $\mathbb{P} = \mathbb{P}^N$ such that

$$L = \mathcal{O}_X(1) \coloneqq \mathcal{O}_{\mathbb{P}^N}(1) \mid X.$$

(ii). L is ample if $L^{\otimes m}$ is very ample for some $m > 0$.

A Cartier divisor D on X is ample or very ample if the corresponding line bundle $\mathcal{O}_X(D)$ is so.

Remark 1.2 (Amplitude)**.** We will interchangeably use "ampleness" and "amplitude" to describe the property of being ample.

Example 1.3 (Ample line bundle on curves)**.** If X is an irreducible curve, and L is a line bundle on X, then L is ample if and only if $\deg(L) > 0$. See [SP26, Tag 0B5X].

Example 1.4 (Varieties with $\mathrm{Pic} = \mathbb{Z}$)**.** If X is a projective variety with $\mathrm{Pic}(X) = \mathbb{Z}$, then any non-zero effective divisor on X is ample. This applies, for instance, when X is a projective space or a Grassmannian.

Proof. Let A be the generator of $\mathrm{Pic}(X)$, then $\mathcal{O}_X(1) = A^{\otimes m}$ very ample for some m. Replace A to A^{-1} if necessary, may assume $m > 0$. So $A^{\otimes m}$ very ample, A is ample. Let D be a nonzero Cartier divisor with $\mathcal{O}_X(D) = A^{\otimes n}$, it only suffices to prove $n > 0$.

If $n < 0$, $\mathcal{O}_X(-D) = A^{\otimes(-n)}$ is ample, by Theorem 1.6 (iv) there exists r sufficiently large such that $\mathcal{O}_X(-rD)$ is very ample, so is generated by global sections. However, rD is effective, so $\mathcal{O}_X(-rD)$ is the ideal sheaf defining the associated locally principal closed subscheme, get

$H^0(X, \mathcal{O}_X(-rD)) \hookrightarrow H^0(X, \mathcal{O}_X) = k$ by [Har77, I, 3.4]. For $0 \neq s \in H^0(X, \mathcal{O}_X(-rD)) \subset k$, s must vanish on $\mathrm{supp}(rD)$, a contradiction. Therefore $H^0(X, \mathcal{O}_X(-rD)) = 0$, contradicts to $\mathcal{O}_X(-rD)$ has enough global sections. Similarly $n \neq 0$. Hence we conclude $n > 0$. □

Example 1.5 (Intersection products)**.** If $D_1, \cdots, D_n$ are ample divisors on an n-dimensional projective variety X, then $(D_1 \cdots D_n) > 0$.

Proof. We recall [dFEM14, Remark 1.2.7]. Let $L_1, \cdots, L_s$ are ample line bundles on X for some $s \leq n$. In this case, there is a positive integer m and a closed subscheme Y of X of dimension $n - s$ such that

$$(L_1 \cdots L_s \cdot L'_1 \cdots L'_{n-s}) = \frac{1}{m}(L'_1 \cdots L'_{n-s} \cdot Y)$$

for every $L'_1, \cdots, L'_{n-s} \in \mathrm{Pic}(X)$. Indeed, if m_i is a positive integer, for $1 \leq i \leq s$, such that $L_i^{m_i}$ is very ample and if $D_i \in |L_i^{m_i}|$ is a general element, then the closed subscheme $Y = D_1 \cap \cdots \cap D_s$ has dimension $n - s$ by Projective Dimension Theorem ([Har77, I, 7.2]). A repeated application of [Deb01, Proposition 1.8] gives the equality, with $m = \prod m_i$.

Since D_i ample implies that $m_i D_i$ is very ample for every i. and intersection number is multilinear, so enough to assume D_i are very ample. Then

$$(D_1 \cdots D_n) = \sum_C (D_n \cdot C)$$

where C runs over the irreducible components. We reduces to Example 1.3 (for D_n is ample on C we gonna to prove later). □

1.1 Cohomological Properties

The first basic fact is that amplitude can be detected cohomologically:

Theorem 1.6 (Cartan–Serre–Grothendieck theorem)**.** Let L be a line bundle on a complete scheme X. The following are equivalent:

(i). L is ample.

(ii). [SP26, Tag 0B5T] Given any coherent sheaf $\mathcal{F}$ on X, there exists a positive integer $m_1 = m_1(\mathcal{F})$ having the property that

$$H^i\left(X, \mathcal{F} \otimes L^{\otimes m}\right) = 0$$

for all $i > 0$, $m \geq m_1(\mathcal{F})$.

(iii). [SP26, Tag 01Q3] Given any coherent sheaf $\mathcal{F}$ on X, there exists a positive integer $m_2 = m_2(\mathcal{F})$ such that $\mathcal{F} \otimes L^{\otimes m}$ is generated by its global sections for all $m \geq m_2(\mathcal{F})$.

(iv). There is a positive integer $m_3 > 0$ such that $L^{\otimes m}$ is very ample for every $m \geq m_3$.

Remark 1.7 (Serre vanishing)**.** The conclusion in (ii) is often referred to as Serre's vanishing theorem.

Remark 1.8 (Amplitude on non-complete schemes)**.** One can also discuss ample line bundles on possibly non-complete schemes. In this more general setting, property (iii) from Theorem 1.6 is taken as the definition of amplitude.

Example 1.9. Let B be a globally generated line bundle on a complete variety or scheme X. Then

$$U \coloneqq \{y \in X \mid B \otimes \mathcal{I}_y \text{ is globally generated}\}$$

is an open subset of X.

Proof. I don't know how to prove the assertion following the proof in [Laz04]. But here I will provide another proof thanks to GPT 5.6 Sol. We claim that: if $p\colon Z \to S$ proper, $\alpha\colon \mathcal{F} \to \mathcal{G}$ is a morphism between coherent sheaves. Then

$$\left\{s \in S \mid \alpha|_{Z_s} \text{ surjective}\right\} = S \setminus p\left(\operatorname{supp}(\operatorname{Coker}\alpha)\right)$$

is open.

The exact sequence

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \longrightarrow \operatorname{Coker}\alpha \longrightarrow 0$$

induces exact sequence

$$\mathcal{F}|_{Z_s} \xrightarrow{\alpha|_{Z_s}} \mathcal{G}|_{Z_s} \longrightarrow \operatorname{Coker}\alpha|_{Z_s} \longrightarrow 0\ .$$

So $\alpha|_{Z_s}$ surjective if and only if $\operatorname{Coker}\alpha|_{Z_s} = 0$.

We prove $\operatorname{Coker}\alpha|_{Z_s} = 0$ if and only if $Z_s \cap \operatorname{supp}(\operatorname{Coker}\alpha) = \emptyset$. $Z_s \cap \operatorname{supp}(\operatorname{Coker}\alpha) = \emptyset$ clearly implies that $\operatorname{Coker}\alpha|_{Z_s} = 0$. For the converse, choose arbitrary $z \in Z_s$, we have local homomorphism $\mathcal{O}_{S,s} \to \mathcal{O}_{Z,z}$. Since $\mathcal{O}_{Z_s,z} = \mathcal{O}_{Z,z}/\mathfrak{m}_s\mathcal{O}_{Z,z}$,

$$\left(\operatorname{Coker}\alpha|_{Z_s}\right)_z \cong (\operatorname{Coker}\alpha)_z \otimes_{\mathcal{O}_{Z,z}} \mathcal{O}_{Z_s,z} \cong \frac{(\operatorname{Coker}\alpha)_z}{\mathfrak{m}_s\mathcal{O}_{Z,z}\,(\operatorname{Coker}\alpha)_z}.$$

This implies that $(\operatorname{Coker}\alpha)_z = \mathfrak{m}_s\mathcal{O}_{Z,z}\,(\operatorname{Coker}\alpha)_z$. Note that $\mathfrak{m}_s\mathcal{O}_{Z,z} \subset \mathfrak{m}_z$, $(\operatorname{Coker}\alpha)_z = 0$ by Nakayama's lemma. Hence $Z_s \cap \operatorname{supp}(\operatorname{Coker}\alpha) = \emptyset$.

Consequently $\alpha|_{Z_s}$ surjective if and only if $Z_s \cap \operatorname{supp}(\operatorname{Coker}\alpha) = \emptyset$ if and only if $s \notin p(\operatorname{supp}(\operatorname{Coker}\alpha))$. By [SP26, Tag 056J], $\operatorname{supp}(\operatorname{Coker}\alpha)$ is closed, so $p(\operatorname{supp}(\operatorname{Coker}\alpha))$ is closed since p is proper.

Now come back to this example. Since B is generated by global sections, we have short exact sequence

$$0 \longrightarrow \mathcal{E} \longrightarrow H^0(X,B) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow B \longrightarrow 0$$

where $\mathcal{E} \coloneqq \operatorname{Ker}\left(H^0(X,B) \otimes_{\mathbb{C}} \mathcal{O}_X \to B\right)$ is coherent. Fix closed point $y \in X$. We have short exact sequence

$$0 \longrightarrow \mathcal{E}_y \longrightarrow H^0(X,B) \otimes_{\mathbb{C}} \mathcal{O}_{X,y} \longrightarrow B_y \longrightarrow 0 \ .$$

Note that B_y is free $\mathcal{O}_{X,y}$-module, so is flat, which induces short exact sequence

$$0 \longrightarrow \mathcal{E}_y/\mathfrak{m}_y\mathcal{E}_y \longrightarrow H^0(X,B) \longrightarrow B_y/\mathfrak{m}_y B_y \longrightarrow 0$$

by tensoring $k(y) = \mathbb{C}$. Compare with

$$0 \longrightarrow H^0\left(X, B \otimes \mathcal{I}_y\right) \longrightarrow H^0(X,B) \longrightarrow B_y/\mathfrak{m}_y B_y \longrightarrow 0 \ ,$$

we get

$$\mathcal{E}_y/\mathfrak{m}_y\mathcal{E}_y \cong H^0(X, B \otimes \mathcal{I}_y).$$

Let $\mathrm{pr}_1, \mathrm{pr}_2 \colon X \times X \to X$ be projections, $\Delta \subset X \times X$ be diagonal image with ideal sheaf $\mathcal{I}_\Delta$. Pull back $\mathcal{E} \to H^0(X,B) \otimes \mathcal{O}_X$ along pr_2, we obtain $\mathrm{pr}_2^*\mathcal{E} \to H^0(X,B) \otimes \mathcal{O}_{X\times X}$; pull back $H^0(X,B) \otimes_{\mathbb{C}} \mathcal{O}_X \to B$ along pr_1, we obtain $H^0(X,B) \otimes \mathcal{O}_{X\times X} \to \mathrm{pr}_1^* B$. Composite, get $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B$. Since $\mathrm{pr}_1^* B$ locally free, we have short exact sequence

$$0 \longrightarrow \mathrm{pr}_1^* B \otimes \mathcal{I}_\Delta \longrightarrow \mathrm{pr}_1^* B \longrightarrow \mathrm{pr}_1^* B \otimes \mathcal{O}_\Delta \longrightarrow 0 \ .$$

To show $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B$ factors through $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B \otimes \mathcal{I}_\Delta \hookrightarrow \mathrm{pr}_1^* B$, it suffices to show $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B \to \mathrm{pr}_1^* B \otimes \mathcal{O}_\Delta$ is zero. Note that $\mathrm{pr}_1|_\Delta = \mathrm{pr}_2|_\Delta = \mathrm{id}_X$, by restricting to Δ, $\mathcal{E} \to H^0(X,B) \otimes \mathcal{O}_X \to B$ is zero, so $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B \to \mathrm{pr}_1^* B \otimes \mathcal{O}_\Delta$ is zero since the target is supported on Δ.

Consider $\mathrm{pr}_2 \colon X \times X \to X$, the fiber of y is $X \times \{y\} \cong X$. Denote $i_y \colon X \times \{y\} \to X \times X$ be inclusion. On the one hand, $i_y^* \mathrm{pr}_2^*\mathcal{E} \cong (\mathcal{E} \times k(y)) \otimes_{\mathbb{C}} \mathcal{O}_X \cong H^0\left(X, B \otimes \mathcal{I}_y\right) \otimes_{\mathbb{C}} \mathcal{O}_X$. On the other hand, clearly $i_y^* \mathrm{pr}_1^* B \cong B$. By $\mathrm{pr}_2|_\Delta : \Delta \to X$ is isomorphism, $\mathcal{O}_\Delta$ is flat, we get short exact sequence

$$0 \longrightarrow i_y^*\mathcal{I}_\Delta \longrightarrow \mathcal{O}_{X\times\{y\}} \cong \mathcal{O}_X \longrightarrow \mathcal{O}_{\Delta\times \operatorname{Spec} k(y)} \cong k(y) \longrightarrow 0 \ .$$

This implies that $i_y^*\mathcal{I}_\Delta \cong \mathcal{I}_y$. On the fiber $\mathrm{pr}_2^*\mathcal{E} \to \mathrm{pr}_1^* B \otimes \mathcal{I}_\Delta$ becomes $H^0(X, B\otimes\mathcal{I}_y)\otimes_{\mathbb{C}}\mathcal{O}_X \to B\otimes\mathcal{I}_y$. It is precisely the evaluation map.

From our above claim, U is open. □

Example 1.10 (Sums of divisors)**.** Let D and E be (Cartier) divisors on a projective scheme X. If D is ample, then there exists m_1 such that mD is very ample for $m \geq m_1$, and there exists m_2 such that $mD+E$ is generated by global sections for $m \geq m_2$. Using [Har77, II, Exercise 7.5 (d)], $mD+E$ is very ample for $m \geq m_1+m_2$. Hence $mD+E$ is generated by global sections.

Example 1.11 (Ample line bundles on a product)**.** If L and M are ample line bundles on projective schemes X and Y respectively, then $\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M$ is ample on $X\times Y$.

Proof. We pick N sufficiently large such that $L^{\otimes N}$ is very ample on X and $M^{\otimes N}$ is very ample on Y by [Har77, II, Exercise 7.5 (e)]. It follows that

$$L^{\otimes N} \cong i_1^*\mathcal{O}_{\mathbb{P}^r}(1), \quad i_1 \colon X \hookrightarrow \mathbb{P}^r,$$
$$M^{\otimes N} \cong i_2^*\mathcal{O}_{\mathbb{P}^s}(1), \quad i_2 \colon Y \hookrightarrow \mathbb{P}^s.$$

Then

$$\begin{aligned}\mathrm{pr}_1^* L^{\otimes N} \otimes \mathrm{pr}_2^* M^{\otimes N} &\cong (i_1\circ\mathrm{pr}_1)^*\mathcal{O}_{\mathbb{P}^r}(1) \otimes (i_2\circ\mathrm{pr}_2)^*\mathcal{O}_{\mathbb{P}^s}(1)\\ &\cong \mathcal{O}_{X\times Y}(1) && \text{([Har77, II, Exercise 5.12])}\\ &= i^*\mathcal{O}_{\mathbb{P}^{rs+r+s}}(1)\end{aligned}$$

where $i\colon X\times Y \hookrightarrow \mathbb{P}^r\times\mathbb{P}^s \hookrightarrow \mathbb{P}^{rs+r+s}$ is Segre embedding. So $\mathrm{pr}_1^* L^{\otimes N} \otimes \mathrm{pr}_2^* M^{\otimes N} \cong (\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M)^{\otimes N}$ is ample on $X\times Y$. By [Har77, II, 7.6] $\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M$ is ample on $X\times Y$. □

Remark 1.12 (Matsusaka's theorem)**.** According to Theorem 1.6, if L is an ample line bundle on a projective variety X then there is an integer $m(L)$ such that $L^{\otimes m}$ is very ample for $m \geq m(L)$. However, the proof of the theorem fails to give any concrete information about the value of this integer. So it is interesting to ask what geometric information $m(L)$ depends on, and whether one can give effective estimates. A theorem of Matsusaka [Mat72] and Kollár–Matsusaka [KMa83] states that if X is a smooth projective variety of dimension n, then one can find $m(L)$ depending only on the intersection numbers (L^n) and $(L^{n-1}\cdot K_X)$. Siu [Siu93] used the theory of multiplier ideals to give an effective statement, which was subsequently improved and clarified by Demailly [Dem96].

Proposition 1.13 (Finite pullbacks, I)**.** Let $f\colon Y \to X$ be a finite mapping of complete schemes, and L an ample line bundle on X. Then f^*L is an ample line bundle on Y. In particular, if $Y \subset X$ is a subscheme of X, then the restriction $L \mid Y$ of L to Y is ample.

Remark 1.14. See Corollary 1.28 for a partial converse.

Proof. Let $\mathcal{F}$ be a coherent sheaf on Y. Then by projection formula ([Har77, II, Exercise 5.1 (d)]) we have $f_*\left(\mathcal{F} \otimes f^*L^{\otimes m}\right) = f_*\mathcal{F} \otimes L^{\otimes m}$, and $f_*\mathcal{F}$ is coherent by [Har77, II, Exercise 5.5 (c)]. And since f is a finite map, it is an affine map, so using [Har77, III, Exercise 8.2], for $m \gg 0$,

$$\begin{aligned} H^i\left(Y, \mathcal{F} \otimes f^*L^{\otimes m}\right) &\cong H^i\left(X, f_*\left(\mathcal{F} \otimes f^*L^{\otimes m}\right)\right) \\ &\cong H^i\left(X, f_*\mathcal{F} \otimes L^{\otimes m}\right) \\ &= 0 \end{aligned}$$

by L is ample on X. So f^*L is ample on Y.

In particular when $f\colon Y \hookrightarrow X$ is closed embedding, $f^*L = L \otimes \mathcal{O}_Y = L \mid Y$. □

Corollary 1.15 (Globally generated line bundles)**.** Suppose that L is globally generated, and let

$$\phi = \phi_{|L|}\colon X \longrightarrow \mathbb{P} = \mathbb{P}H^0(X, L)$$

be the resulting map to projective space defined by the complete linear system $|L|$. Then the following are equivalent:

(i). L is ample.

(ii). ϕ is finite map.

(iii). $(L \cdot C) = \deg(L \mid C) > 0$ for every irreducible curve $C \subset X$.

Proof. (ii) $\Longrightarrow$ (i): ϕ is finite, and $\mathcal{O}_{\mathbb{P}}(1)$ is ample on $\mathbb{P}$, so $L \cong \phi^*\mathcal{O}_{\mathbb{P}}(1)$ is ample by the previous proposition.

(i) $\Longrightarrow$ (ii): Suppose ϕ is not a finite map, then there is a subvariety $Z \subset X$ of $\dim > 0$ that contracted by ϕ to a closed point p, since by [Har77, III, Exercise 11.2] we have under projective map ϕ, quasi-finite is equivalent to finite. Then

$$\begin{aligned} L \mid Z &\cong \left(\phi^*\mathcal{O}_{\mathbb{P}}(1)\right) \mid Z \\ &= \left(\phi|_Z\right)^*\left(\mathcal{O}_{\mathbb{P}}(1) \mid \{p\}\right) \\ &\cong \left(\phi|_Z\right)^*\mathcal{O}_p \\ &\cong \mathcal{O}_Z \end{aligned}$$

is a trivial line bundle, and $L \mid Z$ not ample, so is L by the previous proposition.

(i) $\Longrightarrow$ (iii): L is ample on X implies that $L \mid C$ is ample on C, so $\deg(L \mid C) > 0$ by [Har77, IV, 3.3].

(iii) $\Longrightarrow$ (ii): Also as above where ϕ is not finite, then for $C \subset Z$ irreducible curve, we have

$$\deg(L \mid C) = \deg \mathcal{O}_C = 0.$$

□

The next result allows one in practice to restrict attention to reduced and irreducible varieties.

Proposition 1.16. Let X be a complete scheme, and L a line bundle on X.

(i). L is ample on X if and only if L_{red} is ample on X_{red}.

(ii). L is ample on X if and only if the restriction of L to each irreducible component of X is ample.

Proof. (i). Since $i\colon X_{\text{red}} \hookrightarrow X$ is finite, L is ample implies that $L_{\text{red}} = i^*L$ is ample on X_{red}.

Assume L_{red} is ample on X_{red}. Fix coherent sheaf $\mathcal{F}$ on X, and let $\mathcal{N}$ be the nilradical ideal sheaf of $\mathcal{O}_X$ such that $\mathcal{N}^r = 0$ for some r. Consider the filtration

$$\mathcal{F} \supset \mathcal{N} \cdot \mathcal{F} \supset \mathcal{N}^2 \cdot \mathcal{F} \supset \cdots \supset \mathcal{N}^r \cdot \mathcal{F} = 0,$$

where the quotients $\mathcal{N}^i\mathcal{F}/\mathcal{N}^{i+1}\mathcal{F} \cong \mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X/\mathcal{N} \cong \mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}} = i^*\left(\mathcal{N}^i\mathcal{F}\right)$ are coherent $\mathcal{O}_{X_{\text{red}}}$-modules.

For $i\colon X_{\text{red}} \hookrightarrow X$ closed immersion,

$$\begin{aligned}
H^j\left(X, \left(\mathcal{N}^i\mathcal{F}/\mathcal{N}^{i+1}\mathcal{F}\right) \otimes_{\mathcal{O}_X} L^{\otimes m}\right) &\cong H^j\left(X, \left(\mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}}\right) \otimes_{\mathcal{O}_X} L^{\otimes m}\right) \\
&\cong H^j\left(X, \left(\mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}}\right) \otimes_{\mathcal{O}_{X_{\text{red}}}} L^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}}\right) \\
&\cong H^j\left(X, i_*\left(\mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}} \otimes_{\mathcal{O}_{X_{\text{red}}}} L_{\text{red}}^{\otimes m}\right)\right) \\
\text{[Har77, III, 2.10]} \quad &\cong H^j\left(X_{\text{red}}, \left(\mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}}\right) \otimes_{\mathcal{O}_{X_{\text{red}}}} L_{\text{red}}^{\otimes m}\right) \\
&= 0
\end{aligned}$$

by L_{red} is ample on X_{red}, for $j > 0$ and $m \gg 0$.

Now twisting the exact sequence

$$0 \longrightarrow \mathcal{N}^{i+1}\mathcal{F} \longrightarrow \mathcal{N}^i\mathcal{F} \longrightarrow \mathcal{N}^i\mathcal{F}/\mathcal{N}^{i+1}\mathcal{F} \longrightarrow 0$$

by $L^{\otimes m}$ and taking cohomology, induce the long exact sequence

$$\longrightarrow H^j\left(X, \mathcal{N}^{i+1}\mathcal{F}\otimes L^{\otimes m}\right) \longrightarrow H^j\left(X, \mathcal{N}^{i}\mathcal{F}\otimes L^{\otimes m}\right) \longrightarrow H^j\left(X, \left(\mathcal{N}^{i}\mathcal{F}/\mathcal{N}^{i+1}\mathcal{F}\right)\otimes L^{\otimes m}\right) \longrightarrow$$

$\mathcal{N}^i$ vanishes for i sufficiently large, so cohomologies vanishes for all j, and so we can use decreasing induction on i: the left term is 0 by induction hypothesis, right term is 0 by above computation, so the middle term is vanished. Hence

$$H^j\left(X, \mathcal{N}^{i}\mathcal{F}\otimes L^{\otimes m}\right) = 0 \text{ for } j>0 \text{ and } m \gg 0$$

holds for every i, and when $i=0$ is just telling the ampleness of L.

(ii). We have already proved that ampleness is same on the reduced scheme structure, so can just assume X is reduced.

If L is ample, the restriction of L to each irreducible component of X is ample by the previous proposition.

Let $X = X_1 \cup \cdots \cup X_r$ for X_i irreducible components and assume $L \mid X_i$ is ample for all i. Fix coherent sheaf $\mathcal{F}$ on X, let $\mathcal{I}$ be the ideal sheaf of X_1 in $\mathcal{O}_X$, and consider the exact sequence

$$0 \longrightarrow \mathcal{I}\cdot\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{I}\cdot\mathcal{F} \longrightarrow 0\ .$$

The first term is supported on $X_2 \cup \cdots \cup X_r$ since vanished on X_1, and last term is on X_1 since $\mathcal{F}/\mathcal{I}\mathcal{F} \cong \mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_X/\mathcal{I}$ such that $\operatorname{supp}(\mathcal{F}\otimes\mathcal{O}_X/\mathcal{I}) \subset \operatorname{supp}(\mathcal{O}_X/\mathcal{I}) = X_1$.

So by induction on the number of irreducible components, get that

$$H^j\left(X, \mathcal{I}\mathcal{F}\otimes L^{\otimes m}\right) = H^j\left(X, (\mathcal{F}/\mathcal{I}\mathcal{F})\otimes L^{\otimes m}\right) = 0$$

for $j>0$ and $m \gg 0$, then the middle term $H^j(X, \mathcal{F}\otimes L^{\otimes m}) = 0$ for $j>0$ and $m \gg 0$, implies that L is ample on X.

□

The next theorem shows that—in an extremely strong sense—amplitude is an open condition in families.

Theorem 1.17 (Amplitude in families)**.** Let $f\colon X \to T$ be a proper morphism of schemes, and L a line bundle on X. Given $t \in T$, write

$$X_t = X \times_T \operatorname{Spec} k(t) = f^{-1}(t), \quad L_t = L \mid X_t = L \otimes_{\mathcal{O}_X} \mathcal{O}_{X_t}.$$

Assume that L_0 is ample on X_0 for some point $0 \in T$. Then there is an open neighborhood U of 0 in T such that L_t is ample on X_t for all $t \in U$.

Proof. The statement is local on T, so can assume $T = \operatorname{Spec} A$ is affine.

For arbitrary fixed coherent sheaf $\mathcal{F}$ on X, we claimed that there is a positive integer $m(\mathcal{F})$ such that

$$R^i f_* \left(\mathcal{F} \otimes L^{\otimes m}\right) = 0 \text{ in a neighborhood } U_m \subset T \text{ of } 0 \tag{1}$$

for all $i \geq 1$ and $m \geq m(\mathcal{F})$, which $R^i f_* \left(\mathcal{F} \otimes L^{\otimes m}\right) \cong H^i \left(X, \mathcal{F} \otimes L^{\otimes m}\right)^{\sim}$ by [Har77, III, 8.5].

In fact, by Grothendieck vanishing theorem, this is true for sufficient large i, so we proved by decreasing induction on i. Assuming that equation is known for given $i \geq 2$, need to show that is holds for $i - 1$.

Consider the maximal ideal $\mathfrak{m}_0 \subset A$ of 0 in T, and choose generators $u_1, \cdots, u_p \in \mathfrak{m}_0$. This give rise to a presentation

$$A^{\oplus p} \xrightarrow{u} A \longrightarrow A/\mathfrak{m}_0 = k(0) \longrightarrow 0$$

of $A/\mathfrak{m}_0$, where $u(a_1, \cdots, a_p) = \sum a_i u_i$, then applying $\widetilde{(-)}$,

$$\mathcal{O}_T^{\oplus p} \longrightarrow \mathcal{O}_T \longrightarrow \mathcal{O}_T/\mathfrak{m}_0\mathcal{O}_T \longrightarrow 0 \ .$$

Pull back to f,

$$f^*\mathcal{O}_T^{\oplus p} \longrightarrow f^*\mathcal{O}_T \longrightarrow f^*\left(\mathcal{O}_T/\mathfrak{m}_0\mathcal{O}_T\right) \longrightarrow 0 \ ,$$

which is precisely

$$\mathcal{O}_X^{\oplus p} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{X_0} \longrightarrow 0$$

since $f^*\left(\mathcal{O}_T/\mathfrak{m}_0\right) \cong \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_T} f^{-1}\mathcal{O}_T/f^{-1}\mathfrak{m}_0 \cong \mathcal{O}_X/\mathfrak{m}_0\mathcal{O}_X = \mathcal{O}_{X_0}$.

Tensoring with $\mathcal{F}$ we arrive at an exact diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \operatorname{Ker}\left(f^*u \otimes 1\right) & \longrightarrow & \mathcal{O}_X^{\oplus p} \otimes \mathcal{F} & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F} \otimes \mathcal{O}_{X_0} \longrightarrow 0 \\ & & & & & \searrow & \uparrow & & \\ & & & & & & \operatorname{Im}\left(f^*u \otimes 1\right) & & \\ & & & & & \nearrow & \searrow & & \\ & & & & 0 & & & 0 & \end{array}$$

By the induction hypothesis applied to the kernel sheaf,

$$R^i f_* \left(\operatorname{Ker}\left(f^* \otimes 1\right) \otimes L^{\otimes m}\right) = 0 \text{ near } 0 \text{ for } m \gg 0.$$

Furthermore, since L_0 is ample, the index $i-1$ higher direct image of $\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_{X_0}\otimes_{\mathcal{O}_X}L^{\otimes m}\cong \mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_{X_0}\otimes_{\mathcal{O}_{X_0}}L_0^{\otimes m}$ vanish when m sufficiently large by Serre vanishing, which is just the cohomology group on X_0, follows from [Har77, III, 2.10 and 8.5]:

$$\begin{aligned} R^{i-1}f_*\left(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_{X_0}\otimes_{\mathcal{O}_X}L^{\otimes m}\right) &= H^{i-1}\left(X,\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_{X_0}\otimes_{\mathcal{O}_X}L^{\otimes m}\right)^{\sim} \\ &= H^{i-1}\left(X_0,\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{O}_{X_0}\otimes_{\mathcal{O}_{X_0}}L_0^{\otimes m}\right)^{\sim} \\ &= 0. \end{aligned}$$

Then we get morphisms after tensoring $L^{\otimes m}$ and chasing the above diagram:

$$R^{i-1}f_*\left(\mathcal{O}_X^p\otimes\mathcal{F}\otimes L^{\otimes m}\right)\longrightarrow R^{i-1}f_*\left(\operatorname{Im}\left(f^*u\otimes 1\right)\otimes L^{\otimes m}\right)\longrightarrow R^if_*\left(\operatorname{Ker}\left(f^*u\otimes 1\right)\otimes L^{\otimes m}\right)\ ,$$

$$R^{i-1}f_*\left(\operatorname{Im}\left(f^*u\otimes 1\right)\otimes L^{\otimes m}\right)\longrightarrow R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)\longrightarrow R^{i-1}f_*\left(\mathcal{F}\otimes\mathcal{O}_{X_0}\otimes L^{\otimes m}\right)=0\ .$$

Combine with projection formula [Har77, III, Exercise 8.3], we get

$$R^{i-1}f_*\left(\mathcal{O}_X^p\otimes_{\mathcal{O}_X}\mathcal{F}\otimes L^{\otimes m}\right)\cong R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)\otimes_{\mathcal{O}_T}\mathcal{O}_T^p\xrightarrow{1\otimes u}R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right),$$

this is surjective in a neighborhood U_m' of 0 for $m\gg 0$.

On the other hand, by our construction $1\otimes u$ factor through the inclusion

$$\begin{array}{ccc} R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)\otimes_{\mathcal{O}_T}\mathcal{O}_T^p & \xrightarrow{\qquad 1\otimes u\qquad} & R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right) \\ & \searrow \quad \nearrow & \\ & \mathfrak{m}_0\cdot R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right) & \end{array}$$

since $u_1,\cdots,u_p$ generates $\mathfrak{m}_0$. So in other words, for m sufficiently large, then

$$R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)=\mathfrak{m}_0\cdot R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)$$

in a neighborhood of 0. By Nakayama's lemma, this implies that

$$R^{i-1}f_*\left(\mathcal{F}\otimes L^{\otimes m}\right)=0$$

near 0, as required. Thus we have verified (1).

Next we claim that the canonical mapping

$$\rho_m\colon f^*f_*L^{\otimes m}\longrightarrow L^{\otimes m}$$

is surjective along X_t for all t in a neighborhood U_m'' of 0 provided that $m\gg 0$.

To see this, apply (1) to the ideal sheaf $\mathcal{I}_{X_0/X}$ of X_0 in $\mathcal{O}_X$, can find that

$$R^if_*\left(\mathcal{I}_{X_0/X}\otimes L^{\otimes m}\right)=0 \text{ in a neighborhood } U_m\subset T \text{ of } 0$$

for all $i \geq 1$, $m \gg 0$, so that

$$0 \longrightarrow \mathcal{I}_{X_0/X} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X/\mathcal{I}_{X_0/X} \cong \mathcal{O}_{X_0} \longrightarrow 0$$

induce to

$$H^0\left(X, L^{\otimes m}\right)^{\sim} \longrightarrow H^0\left(X, L^{\otimes m} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0}\right)^{\sim}$$

is surjective in a neighborhood $U_m \subset T$ of 0, but the right term is just $H^0\left(X_0, L_0^{\otimes m}\right)^{\sim}$ support on a neighborhood of 0, so

$$f_*\left(L^{\otimes m}\right) \longrightarrow f_*\left(L^{\otimes m} \otimes \mathcal{O}_{X_0}\right) = H^0\left(X_0, L_0^{\otimes m}\right)^{\sim} \tag{2}$$

is surjective when $m \gg 0$. But by [Har77, II, 7.5], all sufficiently large powers if the ample line bundle L_0 are globally generated.

Composing (2) with the evaluation map

$$H^0\left(X, L_0^{\otimes m}\right) \otimes_{\mathbb{C}} \mathcal{O}_{X_0} \longrightarrow L_0^{\otimes m}, \quad \sum v_i \otimes f_i \longmapsto \sum f_i \cdot v_i$$

which is also surjective, we get

$$f_*\left(L^{\otimes m}\right) \longrightarrow L_0^{\otimes m}$$

is surjective along X_0, i.e.,

$$f_*L^{\otimes m} \otimes_{\mathcal{O}_T} \mathcal{O}_{X_0} \cong f^*f_*L^{\otimes m} \mid X_0 \longrightarrow L^{\otimes m} \mid X_0 = L_0^{\otimes m}$$

says that ρ_m is surjective along X_0, for $m \gg 0$. This implies that $\mathrm{Coker}(\rho_m) = 0$ along X_0 for $m \gg 0$. By [SP26, Tag 056J] $\mathrm{supp}\left(\mathrm{Coker}(\rho_m)\right)$ is closed in X, so $f\left(\mathrm{supp}\left(\mathrm{Coker}(\rho_m)\right)\right)$ is closed in T. Then $T \setminus f\left(\mathrm{supp}\left(\mathrm{Coker}(\rho_m)\right)\right)$ becomes an open neighborhood of 0, and for arbitrary t in this neighborhood, $X_t \cap \mathrm{supp}\left(\mathrm{Coker}(\rho_m)\right) = \emptyset$, i.e. $\mathrm{Coker}(\rho_m) = 0$ along X_t, as we claimed.

Shrinking T, we just suppose that ρ_m is globally surjective for some fixed $m \gg 0$ (note that all neighborhoods we discussed are about some sufficiently large m, we can not shrinking T and moving m simultaneously), then $f_*\left(L^{\otimes m}\right)$ is also globally generated since T is affine ([Har77, III, 8.5]). So we choosing finitely many sections $\{s_i\}_{i=0}^r \subset H^0\left(T, f_*L^{\otimes m}\right)$ generating $f_*\left(L^{\otimes m}\right)$:

$$\mathcal{O}_T^{r+1} \longrightarrow f_*\left(L^{\otimes m}\right)$$

is surjective, pulling back on X get a surjective map

$$f^*\mathcal{O}_T^{r+1} \longrightarrow f^*f_*\left(L^{\otimes m}\right).$$

Composite with ρ_m, get

$$f^*\mathcal{O}_T^{r+1} \longrightarrow L^{\otimes m}$$

is surjective. This defines a map

$$\phi\colon X \longrightarrow \mathbb{P}\left(\mathcal{O}_T^{r+1}\right) = \mathbb{P}^r \times T$$

over T, where $L^{\otimes m} \cong \phi^*\mathcal{O}_{\mathbb{P}\left(\mathcal{O}_T^{r+1}\right)}(1)$, by [Har77, II, 7.12].

Since L_0 is ample on X_0, ϕ restrict on X_0 is

$$\phi_0 \coloneqq \phi|_{X_0} : X_0 \longrightarrow \mathbb{P}^r \times \{0\} \cong \mathbb{P}^r.$$

Then $L_0^{\otimes m} \cong (\phi_0)^*\mathcal{O}_{\mathbb{P}\left(\mathcal{O}_T^{r+1}\right)}(1) \cong (\phi_0)^*\mathcal{O}_{\mathbb{P}^r}(1)$ says that ϕ_0 is fiber finite, since if $C \subset X_0$ such that $\phi_0(C)$ is a point, then

$$L_0^{\otimes m} \mid C \cong (\phi_0)^*\mathcal{O}_{\mathbb{P}^r}(1) \mid C = \left(\phi_0|_C\right)^* \mathcal{O}_{\mathbb{P}^r}(1) \mid \{\mathrm{pt}\} \cong \mathcal{O}_C$$

is just $\deg\left(L_0^{\otimes m} \mid C\right) = m \cdot \deg(L_0 \mid C) = 0$, so $L_0 \mid C$ is not ample from Example 1.3, neither is L_0 from Proposition 1.13, a contradiction. Hence ϕ_0 is finite, as in [SP26, Tag 02LS]. Hence $\phi_t \coloneqq \phi|_{X_t} : X_t \to \mathbb{P}^r$ is finite for t in a neighborhood of 0 from [SP26, Tag 02OH]. $L_t^{\otimes m} = (\phi_t)^*\mathcal{O}_{\mathbb{P}^r}(1)$ implies that $L_t^{\otimes m}$ is ample by Proposition 1.13, so is L_t. □

Remark 1.18. Observe for later reference that the final step of the proof just completed shows that there is a neighborhood $U = U_m$ of 0 in T such that the mapping

$$\phi\colon X_U \coloneqq f^{-1}(U) \longrightarrow \mathbb{P}^r \times U$$

over U determined by $L^{\otimes m}$ is finite.

We close this discussion by presenting some useful applications of Serre's vanishing.

Example 1.19 (Asymptotic Riemann–Roch on Ample case)**.** Let D be an ample Cartier divisor on an irreducible projective variety X of dimension n. Then

$$h^0\left(X, \mathcal{O}_X(mD)\right) = \frac{(D^n)}{n!} \cdot m^n + O\left(m^{n-1}\right).$$

More generally, if $\mathcal{F}$ is any coherent sheaf on X, then

$$h^0\left(X, \mathcal{F} \otimes \mathcal{O}_X(mD)\right) = \operatorname{rank}(\mathcal{F}) \cdot \frac{(D^n)}{n!} \cdot m^n + O\left(m^{n-1}\right).$$

This immediately follows from Serre's vanishing for ampleness.

Example 1.20 (Upper bounds on h^0)**.** If E is any divisor on an irreducible projective variety X of dimension n, then there is a constant $C > 0$ such that

$$h^0\left(X, \mathcal{O}_X(mE)\right) \leq C \cdot m^n$$

for all m sufficiently large.

Proof. Fix an ample divisor D on X. Then for sufficiently large a, $aD - E$ is an effective divisor, so $\mathcal{O}_X(aD - E)$ has global section. Picking a non-zero section $s \in H^0\left(X, \mathcal{O}_X(aD - E)\right)$, it defines

$$\mathcal{O}_X(mE) \hookrightarrow \mathcal{O}_X(maD), \quad t \longmapsto s^m \cdot t,$$

which is injective since $s \neq 0$ and X integral.

Applying $H^0(X, -)$, get injection

$$H^0(X, \mathcal{O}_X(mE)) \hookrightarrow H^0(X, \mathcal{O}_X(maD)),$$

hence

$$h^0(X, \mathcal{O}_X(mE)) \leq h^0(X, \mathcal{O}_X(maD)) = \frac{(aD)^n}{n!} \cdot m^n + O\left(m^{n-1}\right) \leq C \cdot m^n.$$

□

Example 1.21 (Resolutions of a sheaf)**.** Let X be a projective variety, and D an ample divisor on X. Then any coherent sheaf $\mathcal{F}$ on X admits a (possibly non-terminating) resolution of the form

$$\cdots \longrightarrow \bigoplus \mathcal{O}_X(-p_1 D) \longrightarrow \bigoplus \mathcal{O}_X(-p_0 D) \longrightarrow \mathcal{F} \longrightarrow 0$$

for suitable integers $0 \ll p_0 \ll p_1 \ll \cdots$.

Proof. Compare with [Har77, II, 5.18].

Choose $p_0 \gg 0$ such that $\mathcal{F} \otimes \mathcal{O}_X(p_0 D)$ is globally generated, so has surjective map

$$\bigoplus \mathcal{O}_X \longrightarrow \mathcal{F} \otimes \mathcal{O}_X(p_0 D) \longrightarrow 0.$$

Twisting by $\mathcal{O}_X(-p_0 D)$ then gives rise to a surjection

$$\bigoplus \mathcal{O}_X(-p_0 D) \longrightarrow \mathcal{F} \longrightarrow 0.$$

Then use the same process on the kernel. □

Example 1.22 (Surjectivity of multiplication maps)**.** Let X be a projective variety or scheme, and let D and E be ample Cartier divisors on X. Then there is a positive integer $m_0 = m_0(D, E)$ such that the natural maps

$$H^0\left(X, \mathcal{O}_X(aD)\right) \otimes H^0\left(X, \mathcal{O}_X(bE)\right) \longrightarrow H^0\left(X, \mathcal{O}_X(aD + bE)\right)$$

are surjective whenever $a, b \geq m_0$. More generally, for any coherent sheaves $\mathcal{F}, \mathcal{G}$ on X, there is an integer $m_1 = m_1(D, E, \mathcal{F}, \mathcal{G})$ such that

$$H^0\left(X, \mathcal{F} \otimes \mathcal{O}_X(aD)\right) \otimes H^0\left(X, \mathcal{G} \otimes \mathcal{O}_X(bE)\right) \longrightarrow H^0\left(X, \mathcal{F} \otimes \mathcal{G} \otimes \mathcal{O}_X(aD + bE)\right)$$

is surjective for $a, b \geq m_1$.

Proof. For the first statement consider on $X \times X$ the exact sequence

$$0 \longrightarrow \mathcal{I}_\Delta \longrightarrow \mathcal{O}_{X\times X} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0\ , \tag{3}$$

$\Delta \subset X \times X$ being the diagonal. Writing (aD, bE) for the divisor $\mathrm{pr}_1^*(aD) + \mathrm{pr}_2^*(bE)$ on $X \times X$. Then (D, E) is ample by Example 1.11. By Künneth formula ([SP26, Tag 0BEC]),

$$H^0\left(X \times X, \mathrm{pr}_1^*\,\mathcal{O}_X(aD) \otimes \mathrm{pr}_2^*\,\mathcal{O}_X(bE)\right) \cong H^0(X, \mathcal{O}_X(aD)) \otimes_k H^0(X, \mathcal{O}_X(bE)).$$

On the other hand, let $i\colon \Delta \hookrightarrow X \times X$ be closed immersion,

$$\begin{aligned}
\mathcal{O}_\Delta(aD, bE) &= \mathcal{O}_\Delta \otimes_{\mathcal{O}_{X\times X}} \mathcal{O}_{X\times X}((aD, bE)) \\
&= i_*\mathcal{O}_X \otimes_{\mathcal{O}_{X\times X}} \mathcal{O}_{X\times X}((aD, bE)) \\
&\cong i_*\left(\mathcal{O}_X \otimes_{\mathcal{O}_X} i^*\mathcal{O}_{X\times X}((aD, bE))\right) \\
&\cong i_*\mathcal{O}_X(aD + bE)
\end{aligned}$$

from projection formula [Har77, II, Exercise 5.1 (d)] and

$$\begin{aligned}
i^*\mathcal{O}_{X\times X}((aD, bE)) &\cong i^*\,\mathrm{pr}_1^*\,\mathcal{O}_X(aD) \otimes i^*\,\mathrm{pr}_2^*\,\mathcal{O}_X(bE) \\
&\cong \mathcal{O}_X(aD) \otimes \mathcal{O}_X(bE) \\
&\cong \mathcal{O}_X(aD + bE).
\end{aligned}$$

Hence

$$\begin{aligned}
H^0(X \times X, \mathcal{O}_\Delta((aD, bE))) &\cong H^0(X \times X, i_*\mathcal{O}_X(aD + bE)) \\
&= H^0(X, \mathcal{O}_X(aD + bE)).
\end{aligned}$$

The displayed sequence (3) shows that it suffices to verify that

$$H^1\left(X \times X, \mathcal{I}_\Delta((aD, bE))\right) = 0 \tag{4}$$

for $a, b \geq m_0$. From Example 1.21 there exists resolution

$$\cdots \longrightarrow \bigoplus \mathcal{O}_{X\times X}((-p_1 D, -p_1 E)) \longrightarrow \bigoplus \mathcal{O}_{X\times X}((-p_0 D, -p_0 E)) \longrightarrow \mathcal{I}_\Delta \longrightarrow 0 \ .$$

Tensor $\mathcal{O}_{X\times X}(aD, bE)$, get

$$\cdots \longrightarrow \bigoplus \mathcal{O}_{X\times X}(((a-p_1)D, (b-p_1)E)) \longrightarrow \bigoplus \mathcal{O}_{X\times X}(((a-p_0)D, (b-p_0)E)) \longrightarrow \mathcal{I}_\Delta((aD, bE)) \longrightarrow 0$$

Combine with [Laz04, Proposition B.1.2] and Grothendieck's vanishing it is enough for (4) to produce an integer m_0 such that

$$H^i\left(X \times X, \mathcal{O}_{X\times X}((a-p_{i-1})D, (b-p_{i-1})E)\right) = 0$$

whenever $1 \leq i \leq \dim X \times X$ and $a, b \geq m_0$. Künneth formula gives

$$H^i\left(X \times X, \mathcal{O}_{X\times X}((a-p)D, (b-p)E)\right) \cong \bigoplus_{r+s=i} H^r(X, \mathcal{O}_X((a-p)D)) \otimes_k H^s(X, \mathcal{O}_X((b-p)E)).$$

By Serre's vanishing such m_0 exists.

The second statement is similar. We have natural surjection

$$\mathcal{O}_{X\times X} \otimes_{\mathcal{O}_{X\times X}} \mathrm{pr}_1^* \mathcal{F} \otimes \mathrm{pr}_2^* \mathcal{G} \cong \mathrm{pr}_1^* \mathcal{F} \otimes \mathrm{pr}_2^* \mathcal{G} \twoheadrightarrow \mathcal{O}_\Delta \otimes_{\mathcal{O}_{X\times X}} \mathrm{pr}_1^* \mathcal{F} \otimes \mathrm{pr}_2^* \mathcal{G} \cong \Delta_*(\mathcal{F} \otimes \mathcal{G}).$$

Let $\mathcal{E}$ be the kernel, and use the same process on

$$0 \longrightarrow \mathcal{E} \longrightarrow \mathrm{pr}_1^* \mathcal{F} \otimes \mathrm{pr}_2^* \mathcal{G} \longrightarrow \Delta_*(\mathcal{F} \otimes \mathcal{G}) \longrightarrow 0 \ .$$

□

1.2 Numerical Properties

A second very fundamental fact is that amplitude is characterized numerically:

Theorem 1.23 (Nakai–Moishezon–Kleiman criterion)**.** Let L be a line bundle on a projective scheme X. Then L is ample if and only if

$$\int_V c_1(L)^{\dim(V)} = \left(L^{\dim(V)} \cdot V\right) > 0$$

for every positive-dimensional irreducible subvariety $V \subset X$ (including the irreducible components of X).

Corollary 1.24 (Numerical nature of amplitude)**.** If $D_1, D_2 \in \mathrm{Div}(X)$ are numerically equivalent Cartier divisors on a projective variety or scheme X, then D_1 is ample if and only if D_2 is.

Proof. Due to [Laz04, Lemma 1.1.18], we get that

$$\left(D_1^{\dim(V)} \cdot V\right) = \left(D_2^{\dim(V)} \cdot V\right)$$

are strict larger than 0 simultaneously. □

In particular it makes sense to discuss the amplitude of a class $\delta \in N^1(X)$:

Definition 1.25 (Ample classes)**.** A numerical equivalence class $\delta \in N^1(X)$ is **ample** if it is the class of an ample divisor. Ample (algebraic) classes in $H^2(X, \mathbb{Z})$ or $H^2(X, \mathbb{Q})$ are defined in the same way.

Example 1.26 (Varieties with Picard number 1)**.** If X is a projective variety having Picard number $\rho(X) = 1$, then any non-zero effective divisor on X is ample. This extends Example 1.4, and applies for example to a very general abelian variety having a polarization of fixed type.

Proof. Choose H be an ample divisor. Let $0 \neq D > 0$ be an effective divisor, then there exists $a \in \mathbb{Q}$ such that $D \equiv_{\mathrm{num}} aH$. By Nakai–Moishezon–Kleiman criterion $(H^n) > 0$. On the other hand

$$\begin{aligned}
\left(D \cdot H^{n-1}\right) &= \deg\left(c_1(\mathcal{O}_X(D))c_1(\mathcal{O}_X(H))^{n-1} \cap [X]\right) \\
&= \deg\left(c_1(\mathcal{O}_X(H))^{n-1} \cap (c_1(\mathcal{O}_X(D)) \cap [X])\right) \quad \text{([Ful98, Proposition 2.5])} \\
&= \deg\left(c_1(\mathcal{O}_X(H))^{n-1} \cap [D]\right) \\
&= \sum \mathrm{length}_{\mathcal{O}_{D_i}} \mathcal{O}_D \left(H^{n-1} \cdot D_i\right) \\
&> 0
\end{aligned}$$

by Nakai–Moishezon–Kleiman criterion. So $a > 0$. From Corollary 1.24 D is ample. □

Remark 1.27. The structure of the cone of all ample classes on a fixed projective variety is discussed further in Section 1.4. Several examples are worked out in Section 1.5.

The second corollary shows that amplitude can be tested after pulling back by a finite surjective morphism:

Corollary 1.28 (Finite pullbacks, II)**.** Let $f\colon Y \to X$ be a (finite) surjective mapping of projective schemes, and let L be a line bundle on X. If f^*L is ample on Y, then L is ample on X.

Proof. Let $V \subset X$ be an irreducible variety. Since f is surjective, there must exist a generic point in Y that is mapped to the generic point to V, we pick this one, denote it as η. Taking closure $\overline{\{\eta\}}$ is the irreducible component $W \subset f^{-1}(V)$. But this map $f|_W : W \to V$ might not be of finite.

So we need cutting down the general hyperplanes to ensure quasi-finite. Suppose there is a surjective morphism $W \to V$ projective varieties, say with $\dim(W) = \dim(V) + e$. We claim that we can find W_1 in W of codimension 1 such that $W_1 \to V$ is still surjective (so now $\dim(W_1) = \dim V + (e-1)$). Once we have constructed W_1 we will be done by induction. For this case, just take W_1 be a general hyperplane or hypersurface in W. We can assume that W_1 meets the generic fiber of $W \to V$ in codimension one in that fiber, so $W_1 \to V$ is still surjective. Pictorially, one imagine a threefold mapping to a curve, so the general fiber is a surface. Now think of a divisor in the threefold—as long as you have some choice in choosing the divisor, you can arrange that the divisor meets the original surface fiber in a curve. So you have cut down the relative dimension.

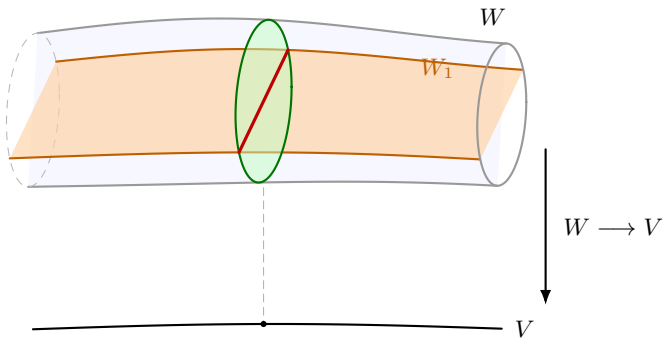


Figure 1: Cut down the general hyperplane.

Here is a more detailed explanation by dimension theory. By [SP26, Tag 0CC1], dominant morphism of integral schemes corresponds to injective ring map on affine open subset, so the generic fiber is also integral. Let ξ (resp. η) be the generic point of W (resp. V). Combine with [SP26, Tag 0A21 (3)] and the proof of [SP26, Tag 0B2L] we have

$$\dim_\xi W_\eta = \dim W_\eta = \dim W - \dim V = e.$$

Choose an embedding $W \hookrightarrow \mathbb{P}^n_k \times_k V = \mathbb{P}^n_V$, take a hyperplane H ($H = H_0 \times_k V$ for hyperplane $H_0 \subset \mathbb{P}^n_k$) with $H \not\supset W$, then $H_\eta \not\supset W_\eta$. Let $W_1 = W \cap H = W \times_{\mathbb{P}^n_V} H$ be the scheme-theoretic

intersection. In $\mathbb{P}^n_{K(V)}$ we have

$$\dim(W_\eta \cap H_\eta) \geq e - 1$$

from [SP26, Tag 0B2R]. On the other hand, $W_\eta \cap H_\eta$ is a proper locally principal closed subscheme, so is an effective Cartier divisor of W_η since W_η is integral. Let ζ be the generic point of one irreducible component of $W_\eta \cap H_\eta$. By [SP26, Tag 0BCN] $\dim \mathcal{O}_{W_\eta,\zeta} = 1$. Since W_η is irreducible, [SP26, Tag 0A21 (3) and (10)] gives

$$e = \dim W_\eta = \dim_\zeta(W_\eta) = \operatorname{trdeg}_{K(V)} k(\zeta) + \dim \mathcal{O}_{W_\eta,\zeta} = \operatorname{trdeg}_{K(V)} k(\zeta) + 1.$$

From [SP26, Tag 0A21 (6)] every such irreducible component has dimension $e - 1$. Therefore

$$\dim(W_\eta \cap H_\eta) = e - 1.$$

W_1 meets the generic fiber of $W \to V$ in codimension one in that fiber. Note that $W_1 \to V$ is dominant. By W_1 proper over k and V separated over k, $W_1 \to V$ is proper from [SP26, Tag 01W6], so $W_1 \to V$ is surjective.

So we get a new map $f|_W : W \to V$ of which we using the same notation. [SP26, Tag 02LS] about proper and quasi-finite is equivalent to finite is what we want the $f\colon W \to V$.

Using the projection formula:

$$\int_W c_1 \left(f^* L\right)^{\dim(W)} = [K(W) : K(V)] \int_V c_1(L)^{\dim(V)} > 0.$$

We conclude that L is ample. □

Proof of Theorem 1.23. Suppose first that L is ample. Then $L^{\otimes m}$ is very ample for some $m \gg 0$, and

$$m^{\dim V} \cdot \int_V (c_1(L))^{\dim V} = \int_V \left(c_1(L^{\otimes m}\right)^{\dim V}$$

is the degree of V in the corresponding projective embedding of X (see [KMo98, Proposition 1.35]. Consequently, this integral is strictly positive.

Assuming that the intersection number is strictly positive. By Proposition 1.16 we can assuming X is irreducible and reduced, i.e., a projective variety. We can associate L to a Cartier divisor D so write $L = \mathcal{O}_X(D)$.

Now we using the induction on $\dim X = n$. The case $n = 1$ is Example 1.3. Assuming assertion holds for dimension $\leq n - 1$, and consider $\dim X = n$.

We assert first that

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \text{ for } m \gg 0.$$

In fact, asymptotic Riemann–Roch gives

$$\chi(X, mD) = \frac{(D^n)}{n!} \cdot m^n + O\left(m^{n-1}\right)$$

and $(D^n) = \int_X c_1(L)^n > 0$ by our condition. By Example 1.10 we can set $A' = mH + D$ and $B = mH$ for $m \gg 0$, both are very ample. Now $H^0(X, \mathcal{O}_X(A'))$ admits nonzero global sections, so $A' \equiv_{\text{lin}} A$ for A effective very ample. Then $D \equiv_{\text{lin}} A - B$ as difference of very ample effective divisors.

We have two exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{O}_X(mD - B) & \xrightarrow{\cdot A} & \mathcal{O}_X((m+1)D) & \longrightarrow & \mathcal{O}_A((m+1)D) & \longrightarrow & 0 \\ & & \| & & & & & & \\ 0 & \longrightarrow & \mathcal{O}_X(mD - B) & \xrightarrow{\cdot B} & \mathcal{O}_X(mD) & \longrightarrow & \mathcal{O}_B(mD) & \longrightarrow & 0 \end{array}$$

where the associated scheme of A and B are strictly smaller dimension than X as they're both effective, so by induction hypothesis, $\mathcal{O}_A(D)$ and $\mathcal{O}_B(D)$ are ample. Consequently the higher cohomology of each of two schemes on the two right terms vanishes when $m \gg 0$. So we have

$$H^i(X, \mathcal{O}_X(mD)) = H^i(X, \mathcal{O}_X(mD - B)) = H^i(X, \mathcal{O}_X((m+1)D))$$

for $i \geq 2$. In other words, for $i \geq 2$, the dimension $h^i(X, mD)$ are eventually constant as m growing. Therefore,

$$\chi(X, \mathcal{O}_X(mD)) = h^0(X, \mathcal{O}_X(mD)) - h^1(X, \mathcal{O}_X(mD)) + C$$

for some constant C and $m \gg 0$. So if $H^0(X, \mathcal{O}_X(mD)) = 0$, then the coefficient of m^n is negative, contradicted with $(D^n) > 0$, we proved what we asserted.

Since D is ample if and only if mD is, there is no loss in generality in replacing D by mD. Therefore we henceforth suppose that D is effective.

Next we show that $\mathcal{O}_X(mD)$ is globally generated for sufficiently large m. Since D is assume to be effective, we do not care about the points away from the associated subscheme, since outside $\text{supp}(D)$, $\mathcal{O}_X(mD) \cong \mathcal{O}_X$ (recall [Har77, II, 6.18]). So the issue is to show that no point of D is a base point of the linear series $|\mathcal{O}_X(mD)|$.

Consider to this end the exact sequence

$$0 \longrightarrow \mathcal{O}_X((m-1)D) \xrightarrow{\cdot D} \mathcal{O}_X(mD) \longrightarrow \mathcal{O}_D(mD) \longrightarrow 0\ . \tag{5}$$

As before, $\mathcal{O}_D(D)$ is ample by induction hypothesis. So $\mathcal{O}_D(mD)$ is globally generated by very ampleness and $H^1(X, \mathcal{O}_D(mD)) = 0$ for $m \gg 0$ by Serre's vanishing. It then follows first of all

from the long exact sequence induced by (5) that the natural homomorphism

$$H^1(X, \mathcal{O}_X((m-1)D)) \longrightarrow H^1(X, \mathcal{O}_X(mD)) \tag{6}$$

is surjective for every $m \gg 0$.

By coherence ([Har77, III, 5.2]), $\dim_{\mathbb{C}} H^1(X, \mathcal{O}_X(mD))$ is finite, so $\dim H^1(X, \mathcal{O}_X(mD))$ become decreasing as m growing, so it must eventually stabilize. That is, the maps in (6) must actually be isomorphisms for sufficiently large m. Therefore the restriction mappings

$$H^0(X, \mathcal{O}_X(mD)) \longrightarrow H^0(X, \mathcal{O}_D(mD)) = H^0(D, \mathcal{O}_D(mD))$$

are surjective for $m \gg 0$.

But since $\mathcal{O}_D(mD)$ is globally generated, by surjectivity sections in left cannot be vanished simultaneously, i.e., no point of $\operatorname{supp}(D)$ is a base point of $|mD|$.

Finally, by Corollary 1.15 and our assumption,

$$\int_C c_1\left(\mathcal{O}_X(mD)\right) = (mD \cdot C) > 0$$

for every irreducible curve $C \subset X$, get that $\mathcal{O}_X(mD)$ is ample, so the $\mathcal{O}_X(D)$. □

Remark 1.29 (Nakai's criterion on proper scheme)**.** The statement of theorem remains true for any complete scheme X, without assuming that X is projective, of which we are using this to give a divisor D as a difference of two very ample divisors. See [Deb01, Theorem 1.21]:

Theorem (Nakai–Moishezon criterion)**.** A Cartier divisor D on a proper scheme X is ample if and only if, for every closed integral subscheme Y of X, one has

$$\left(D^{\dim Y} \cdot Y\right) > 0.$$

Example 1.30 (Divisors with ample normal bundle)**.** We outline a result due to [Har70, III, 4.2] concerning divisors having ample normal bundles. Let X be a projective variety, and let $D \subset X$ be an effective Cartier divisor on X whose normal bundle $\mathcal{O}_D(D)$ is ample. Then:

(i). For $m \gg 0$, $\mathcal{O}_X(mD)$ is globally generated.

(ii). For $m \gg 0$, the restriction

$$H^0(X, \mathcal{O}_X(mD)) \longrightarrow H^0(D, \mathcal{O}_D(mD))$$

is surjective.

(iii). There is a proper birational morphism

$$f\colon X \longrightarrow \overline{X}$$

from X to a projective variety $\overline{X}$ such that f is an isomorphism in a neighborhood of D, and $\overline{D} \coloneqq f(D)$ is an ample effective divisor on $\overline{X}$.

Example 1.31 (Irreducible curves of positive self-intersection on a surface)**.** Let X be a smooth projective surface, and let $C \subset X$ be an irreducible curve with $(C^2) > 0$. Then $\mathcal{O}_X(mC)$ is free, i.e., the complete linear system is base-point free, for $m \gg 0$.

Indeed, by Example 1.3, $(C^2) = \deg \mathcal{O}_C(C) > 0$ implies that $\mathcal{O}_C(C)$ is ample, so the assertion is immediate by Example 1.30.

Example 1.32 (Further characterization of ampleness)**.** Follows from [Kle66, Chapter 3, §1].

Let D be a Cartier divisor on a projective algebraic scheme X. Then D is ample if and only if it satisfies either of the following properties:

(i). For every irreducible subvariety $V \subset X$ of positive dimension, there is a positive integer $m = m(V)$, together with a non-zero section $0 \neq s = s_V \in H^0(V, \mathcal{O}_V(mD))$, such that s vanishes at some point of V.

(ii). For every irreducible subvariety $V \subset X$ of positive dimension,

$$\chi(V, \mathcal{O}_V(mD)) \to \infty \text{ as } m \to \infty.$$

Proof. If D is ample, then $D \mid V$ is ample, so $\mathcal{O}_V(mD)$ is very ample for some $m > 0$. Hence gives an embedding $V \hookrightarrow \mathbb{P}H^0(V, \mathcal{O}_V(mD))$. By the "only if" part of [Har77, II, 7.3] such s in (i) exists.

If D is ample, write $d = \dim V > 0$. Since $D \mid V$ is ample,

$$\chi(V, \mathcal{O}_V(mD)) = \frac{\left(D^d \cdot V\right)}{d!} m^d + O\left(m^{d-1}\right).$$

And ampleness gives $\left(D^d \cdot V\right) > 0$ by Nakai–Moishezon–Kleiman criterion, we conclude that $\chi(V, \mathcal{O}_V(mD)) \to \infty$ as $m \to \infty$.

To prove D is ample, it is enough to prove this when X is reduced and irreducible.

Suppose that (i) holds. As $\mathcal{O}_V(mD)$ has global section, mD is effective on V ([SP26, Tag 01X0]). Take first $V = X$ and replacing D by mD, we can assume D is effective. By induction on dimension, $\mathcal{O}_D(D)$ is ample where D is the associated subscheme. So $\mathcal{O}_X(mD)$ is base

point free for same reason for m sufficiently large, that is, $|mD|$ is free. Consider the morphism $\phi\colon X \to \mathbb{P}H^0(X, \mathcal{O}_X(mD))$. Let $C \subset X$ be an irreducible curve. By (i) there exists $r > 0$ and $0 \neq s \in H^0(C, \mathcal{O}_C(rD))$ such that s vanishes at some point of C, so $\deg \mathcal{O}_C(rD) = r \deg \mathcal{O}_C(D) > 0$ by [SP26, Tag 0B40]. From Corollary 1.15 ϕ is finite, so $\mathcal{O}_X(mD)$ is ample.

Now suppose (ii) holds. By induction one can assume $\mathcal{O}_E(D)$ is ample for every effective divisor E on X. Then the proof of Theorem 1.23 goes through with little change, except that one uses (ii) rather than Riemann–Roch to control $\chi(X, \mathcal{O}_X(mD))$. In order to apply Corollary 1.15, note that

$$\chi(C, \mathcal{O}_C(mD)) = m \deg \mathcal{O}_C(D) + \chi(C, \mathcal{O}_C)$$

(recall [SP26, Tag 0AYR and Tag 0AYX]) forces $\deg \mathcal{O}_C(D) > 0$, $\mathcal{O}_C(D)$ ample, so $\mathcal{O}_X(D)$ restricts to an ample bundle on every irreducible curve in X. So we finish the proof. □

The next example shows that on a scheme of dimension n, dimensions of cohomology groups can grow at most like a polynomial of degree n:

Example 1.33 (Growth of cohomology)**.** Let X be a projective scheme of dimension n and D a divisor on X. If $\mathcal{F}$ is any coherent sheaf on X then

$$h^i(X, \mathcal{F}(mD)) = O(m^n)$$

for every i.

Proof. Write $D \equiv_{\text{lin}} A - B$ as the difference of very ample divisors having the property that neither A nor B contains any of the subvarieties of X defined by the associated primes of $\mathcal{F}$.

Consider two sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}(mD - B) & \xrightarrow{\cdot A} & \mathcal{F}((m+1)D) & \longrightarrow & \mathcal{F} \otimes \mathcal{O}_A((m+1)D) & \longrightarrow & 0 \\ & & \| & & & & & & \\ 0 & \longrightarrow & \mathcal{F}(mD - B) & \xrightarrow{\cdot B} & \mathcal{F}(mD) & \longrightarrow & \mathcal{F} \otimes \mathcal{O}_B(mD) & \longrightarrow & 0 \end{array}$$

are exact by what we setting that $\cdot A$ and $\cdot B$ are indeed injective.

So by induction on dimension n. For $n = 0$, higher cohomologies trivial so $h^i = O(1)$. Suppose the assertion holds for all dimension $< n$. Now for n,

$$\begin{aligned} -O\left(m^{n-1}\right) &= -h^{i-1}\left(A, \mathcal{F}((m+1)D)|_A\right) \\ &\leq h^i(X, \mathcal{F}((m+1)D)) - h^i(X, \mathcal{F}(mD - B)) \\ &\leq h^i\left(A, \mathcal{F}((m+1)D)|_A\right) \\ &= O\left(m^{n-1}\right), \end{aligned}$$

$$\begin{aligned} -O\left(m^{n-1}\right) &= -h^{i-1}\left(B, \mathcal{F}(mD)|_B\right) \\ &\leq h^i(X, \mathcal{F}(mD)) - h^i(X, \mathcal{F}(mD-B)) \\ &\leq h^i\left(B, \mathcal{F}(mD)|_B\right) \\ &= O\left(m^{n-1}\right). \end{aligned}$$

So taking sum together, we get

$$\left|h^i(X, \mathcal{F}((m+1)D)) - h^i(X, \mathcal{F}(mD))\right| = O\left(m^{n-1}\right),$$

that is

$$h^i(X, \mathcal{F}(mD)) = O(m^n).$$

□

Example 1.34. In the setting of above example, it can easily happen that the higher cohomology groups have maximal growth.

For instance, if X is smooth and $-D$ is ample, then $h^n(X, \mathcal{O}_X(mD)) = h^0(X, \mathcal{O}_X(K_X - mD))$ by Serre duality. By Serre vanishing,

$$\chi(X, \mathcal{O}_X(K_X - mD)) = h^0(X, \mathcal{O}_X(K_X - mD)) = \frac{((-D))^n}{n!} m^n + O\left(m^{n-1}\right)$$

grows like m^n.

For a more interesting example, let X be the blowing-up $\mathrm{Bl}_p(\mathbb{P}^2)$ of $\mathbb{P}^2$ at a point, with exceptional divisor E, i.e., $E \cong \mathbb{P}^1$, $\mathcal{O}_E(E) \cong \mathcal{O}_{\mathbb{P}^1}(-1)$. From the short exact sequence

$$0 \longrightarrow \mathcal{O}_X((m-1)E) \longrightarrow \mathcal{O}_X(mE) \longrightarrow \mathcal{O}_E(mE) \longrightarrow 0\ ,$$

combine with $H^0(E, \mathcal{O}_E(mE)) = H^0\left(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-m)\right) = 0$ and $H^2(X, \mathcal{O}_X((m-1)E))^\vee = H^0(X, \mathcal{O}_X(K_X - (m-1)E)) = 0$ (note that $K_X - (m-1)E$ is not effective) we obtain short exact sequence

$$0 \longrightarrow H^1(X, \mathcal{O}_X((m-1)E)) \longrightarrow H^1(X, \mathcal{O}_X(mE)) \longrightarrow H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-m)) \longrightarrow 0\ .$$

It follows that

$$h^1(X, \mathcal{O}_X(mE)) = h^1(X, \mathcal{O}_X((m-1)E)) + (m-1).$$

Hence

$$h^1(X, \mathcal{O}_X(mE)) = \binom{m}{2}.$$

Example 1.35 (Growth of cohomology of pullbacks)**.** Let

$$\mu\colon X' \longrightarrow X$$

be a surjective and generically finite mapping of projective varieties or schemes of dimension n. Fix a divisor D on X and put $D' = \mu^* D$. Then for every $i \geq 0$,

$$h^i(X', \mathcal{O}_{X'}(mD')) = h^i\left(X, (\mu_* \mathcal{O}_{X'}) \otimes \mathcal{O}_X(mD)\right) + O\left(m^{n-1}\right).$$

Proof. By $D' = \mu^* D$,

$$\mathcal{O}_{X'}(mD') = \mathcal{O}_{X'} \otimes \mu^* \mathcal{O}_X(mD).$$

The projection formula gives

$$R^q \mu_* \mathcal{O}_{X'}(mD') \cong R^q \mu_* \mathcal{O}_{X'} \otimes \mathcal{O}_X(mD).$$

The Leray spectral sequence:

$$E_2^{p,q} = H^p\left(X, R^q \mu_* \mathcal{O}_{X'} \otimes \mathcal{O}_X(mD)\right) \Longrightarrow H^{p+q}\left(X', \mathcal{O}_{X'}(mD')\right).$$

For $q = 0$,

$$\dim E_2^{p,0} = h^p\left(X, (\mu_* \mathcal{O}_{X'}) \otimes \mathcal{O}_X(mD)\right).$$

For $q > 0$. Since μ is generically finite, by [Har77, II, Exercise 3.7], there exists open dense subset $U \subset X$ such that $\mu^{-1}(U) \to U$ is finite. Combine with [Har77, III, 3.5 and 8.1],

$$R^q \mu_* \mathcal{O}_{X'}|_U = 0, \quad q > 0.$$

That is, $R^q \mu_* \mathcal{O}_{X'}$ supports on a subset with dimension $\leq n - 1$. Apply Example 1.33 get

$$h^p(X, R^q \mu_* \mathcal{O}_{X'} \otimes \mathcal{O}_X(mD)) = O(m^{n-1}).$$

Therefore,

$$\begin{aligned} h^i(X', \mathcal{O}_{X'}(mD')) &= \sum_{p=0}^{i} \dim E_\infty^{p,i-p} \\ &= \left(\dim E_2^{i,0} + O\left(m^{n-1}\right)\right) + O\left(m^{n-1}\right) \\ &= h^i\left(X, (\mu_* \mathcal{O}_{X'}) \otimes \mathcal{O}_X(mD)\right) + O\left(m^{n-1}\right). \end{aligned}$$

□

Example 1.36 (Higher cohomology of nef divisors, I)**.** Kollár [Kol96, V, 2.15] shows that one can adapt the proof of Nakai's criterion to establish the following.

Theorem. Let X be a projective scheme of dimension n, and D a divisor on X having the property that

$$\left(D^{\dim V} \cdot V\right) \geq 0 \text{ for all irreducible subvarieties } V \subset X.$$

Then

$$h^i\left(X, \mathcal{O}_X(mD)\right) = O\left(m^{n-1}\right) \text{ for } i \geq 1. \tag{7}$$

Remark 1.37 (Complete schemes)**.** With a little more care, one can show that Example 1.33 and Example 1.36 remain valid for arbitrary complete schemes. See [Deb01, Proposition 1.31].

Remark 1.38 (Other ground fields)**.** Except for Matsusaka's theorem (Example 1.12), all of the results and arguments appearing so far in this section remain valid without change for varieties defined over an algebraically closed field of arbitrary characteristic.

References

[Deb01] Olivier Debarre, *Higher-Dimensional Algebraic Geometry*, Universitext, Springer-Verlag, New York, 2001.

[dFEM14] Tommaso de Fernex, Lawrence Ein, and Mircea Mustaţă, *Vanishing Theorems and Singularities in Birational Geometry*, Preprint, 2014, `https://homepages.math.uic.edu/~ein/DFEM.pdf`.

[Dem96] Jean-Pierre Demailly, L^2 vanishing theorems for positive line bundles and adjunction theory, in Fabrizio Catanese and Ciro Ciliberto (eds.), *Transcendental Methods in Algebraic Geometry*, Lecture Notes in Mathematics, Vol. 1646, Springer-Verlag, Berlin–Heidelberg, 1996, 1–97.

[Ful98] William Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, Vol. 2, Springer-Verlag, New York, 1998.

[Har70] Robin Hartshorne, *Ample Subvarieties of Algebraic Varieties*, Notes written in collaboration with C. Musili, Lecture Notes in Mathematics, Vol. 156, Springer-Verlag, Berlin–Heidelberg, 1970.

[Har77] Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, Vol. 52, Springer-Verlag, New York, 1977.

[Kle66] Steven L. Kleiman, Toward a numerical theory of ampleness, *Annals of Mathematics* (2) 84 (1966), no. 3, 293–344.

[Kol96] János Kollár, *Rational Curves on Algebraic Varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, Vol. 32, Springer-Verlag, Berlin, 1996.

[KMa83] János Kollár and Teruhisa Matsusaka, Riemann–Roch type inequalities, *American Journal of Mathematics* 105 (1983), no. 1, 229–252.

[KMo98] János Kollár and Shigefumi Mori, *Birational Geometry of Algebraic Varieties*, Cambridge Tracts in Mathematics, Vol. 134, Cambridge University Press, Cambridge, 1998. With the collaboration of C. H. Clemens and A. Corti.

[Laz04] Robert Lazarsfeld, *Positivity in Algebraic Geometry I: Classical Setting: Line Bundles and Linear Series*, Ergebnisse der Mathematik und ihrer Grenzgebiete, Vol. 48, Springer-Verlag, Berlin–Heidelberg, 2004.

[Mat72] Teruhisa Matsusaka, Polarized varieties with a given Hilbert polynomial, *American Journal of Mathematics* 94 (1972), no. 4, 1027–1077.

[Siu93] Yum-Tong Siu, An effective Matsusaka big theorem, *Annales de l'Institut Fourier* 43 (1993), no. 5, 1387–1405.

[SP26] The Stacks Project Authors, *The Stacks Project*, 2026, `https://stacks.math.columbia.edu/`.